GNBF5020

Practice Examination 01 · Answer Key

Cumulative · Lectures 1–6 · answers, model answers and marking scheme

GNBF5020 · Introduction to Molecular Biology and Genetics

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Keys the paper: GNBF5020_Practice_Exam_01_Cumulative_L1-L6.pdf · 30 marks · 90 minutes

Built strictly from the course materials in this folder: the Lectures 1–5 revision pack, the Lecture 6 guide, the case study bank and the exam focus prediction

Part	Question type	Marks per question	Total
A	Multiple choice — the letter, plus why the distractors fail	2	10
B	Pipeline mapping — one mark per limb, as marked	6	12
C	Exception handling — 3 to find the break, 5 to follow it	8	8
	Total		**30**

[DO NOT OPEN FIRST] Do not open this before you have sat the paper. The question paper is a separate file — `GNBF5020_Practice_Exam_01_Cumulative_L1–L6.pdf`. Sit it closed-book first, then come back here. Reading the key before you attempt it removes exactly the thing the paper is for: finding out which parts of the wiring you actually have.

[HOW TO MARK] How to mark yourself. Every MCQ carries 2 marks: 1 for the letter, 1 for being able to say why it is right **and** why the wrong options fail. A right letter with no reasoning is half marks — because the reasoning is the part the exam actually tests. The pipeline and exception questions are marked limb by limb, and the limb marks are given in the model answers.

[AND AFTER YOU SUBMIT] This key is not the post-mortem. It gives the answers, the model answers and the marks. The full post-mortem — your errors, one by one, each biological mistake translated into computational or data-flow terms — is written when you submit your answers, so that it can address what *you* got wrong rather than what a stranger might.

How to use this key

Marks	What the mark is for
MCQ (2 each)	1 mark for the correct letter, 1 mark for being able to say **why** it is correct and **why each distractor fails**. A right letter with no reason is 1 mark, not 2.
Pipeline (6 each)	1 mark per limb, as marked below. Partial marks are earned by naming the right stage even if the reasoning around it is thin.
Exception (8)	3 marks for locating the break and tracing the consequence, 1 each for the four shorter limbs.

Total: 30 marks.

Everything below is drawn from the same course materials as the paper: the Lectures 1–5 revision pack (audio + slides), the Lecture 6 guide, the case study bank and the exam focus prediction.

PART A — Multiple choice: the key and the reasoning

The correct letter for each question, why it is correct, and — for every option that is wrong — the specific reason it fails. A right letter with no reason is half marks.

Q1 — Answer: **C**

Why C is right. Loss of the Pol ε 3′→5′ exonuclease means misincorporations are no longer caught and excised. The cell keeps dividing with a rising load of spontaneous mutations, which means a rising supply of novel, foreign-looking peptides — **neoantigens** — and therefore a "hot tumour" with cytotoxic T-cell recognition. Checkpoint blockade (anti-PD-1) works by releasing the brake on those T cells, so this is the route into immunotherapy. The course makes the same chain for the POLE ultramutator: mutation burden up → neoantigen up → T-cell infiltration up → the tumour becomes immune-visible.

Note that this route is **independent of MMR status**. The tumour in the stem is microsatellite stable with all four MMR proteins retained, so it will not get the frameshift-driven neoantigen load of an MSI-high tumour. It does not need it — the proofreading defect supplies its own mutations.

Option by option — why each one stands or falls.

A.	fails because PARP inhibition works by trapping PARP on single-strand breaks, converting them into double-strand breaks; the selectivity comes from the tumour already being **HR-deficient (BRCA1/2)** so it cannot clear the damage. A proofreading defect does not abolish homologous recombination, and a misincorporated base is not an unrepaired single-strand break — the drug's mechanism does not even engage this lesion.
B.	fails because this confuses two different jobs. The 3′→5′ exonuclease is the polymerase's *proofreader* for the base it just added. The lesions the ADC payload creates are removed by **BER, NER, HR and NHEJ** — not by polymerase proofreading. The course links proofreading loss to neoantigen load, never to topoisomerase-inhibitor sensitivity.
C.	is right — loss of proofreading raises the spontaneous mutation burden, which generates abundant neoantigens, which makes the tumour immunogenic and responsive to checkpoint blockade. Full reasoning above.
D.	fails because "undruggable" is exactly the word the course retires: K-Ras was historically undruggable and now has an approved targeted inhibitor. And more to the point, the therapy here is not a small molecule aimed at the polymerase — it is immune therapy enabled by the mutation's consequences.
E.	fails because it mixes up two independent routes and then compounds the error. The neoantigen response does **not** require the MSI-high phenotype (proofreading loss generates its own), and PARP inhibition still requires HR deficiency. A regimen built on both assumptions is wrong twice.

In systems terms. The proofreading exonuclease is the **validation layer** of the write pipeline. Remove it and the pipeline does not crash — it keeps running and keeps producing output, just silently corrupted. That is why the phenotype is a mutator phenotype rather than a lethal one, and it is why the right lever is the immune system (the consumer of corrupted

output), not a drug aimed at the corrupted writer.

Q2 — Answer: **B**

Why B is right. The lagging strand is not made as one continuous molecule. It is made as a series of **Okazaki fragments** (primers spaced roughly 200 nt apart), each extended by DNA polymerase, each primer then removed and the gap filled, and then each fragment joined to the next by **DNA ligase**, which seals the nick with a phosphodiester bond. The stem tells you what has *not* failed:

- synthesis is at a normal rate → initiation and extension are fine;
- no RNA is retained in the new DNA → the primers were removed, so primer removal is fine;
- the nicks sit at fragment junctions → the defect is in the step that joins the fragments.

If ligase is missing, every fragment boundary stays an unsealed nick, which is exactly the pattern described. And the leading strand is synthesised continuously as a single piece, so it has no fragment junctions to seal and is essentially unaffected — the strand-specific asymmetry is the giveaway.

Option by option — why each one stands or falls.

A.	fails because without primase the lagging strand cannot be initiated at all: there would be no fragments, and synthesis would not be at a normal rate. The stem shows fragments exist and are joined by something.
B.	is right — the lagging strand is made as discrete Okazaki fragments and each one must be ligated to the next, so a missing ligase leaves a nick at every junction while the continuously synthesised leading strand is untouched. Full reasoning above.
C.	fails because topoisomerase relieves torsional stress ahead of the fork. Its loss stalls or slows the fork and would affect both strands; it does not produce nicks at fragment junctions only on the lagging strand.
D.	fails because loss of proofreading produces **mismatched bases** — substitutions. A mismatch is a wrong letter in a continuous backbone; it is not a break in the backbone. The nicks here are structural, not base-chemistry.
E.	fails because if the RNA primer were never removed, each junction would carry an **RNA–DNA transition with extra nucleotides** — the primer would be copied into the DNA. The stem rules that out explicitly, and the course's own depiction shows primer removal followed by ligation of the resulting nick. The piece that is missing is the ligase.

In systems terms. The fork is a pipeline; the lagging strand is a batch job that emits **fragments** and must concatenate them in a final join step. If the join step is missing, every batch still emits well-formed chunks and the throughput metrics look perfect — but the output is a sequence of separate chunks with a seam at every boundary, and the leading strand (a single streaming write) is untouched. This is the classic failure of *partial* output: everything upstream is healthy, and the defect only appears when someone tries to **parse** the result.

Q3 — Answer: **A**

Why A is right. The question is which version of the assay performs best, and that is decided by **discrimination, not by how much methylation there is**. AUC is the summary of that discrimination across every possible threshold:

Marker	AUC
Exon 1	0.847
Exon 2	0.924
Exon 1 + exon 2	0.938

Exon 2 separates better than exon 1 on its own, and combining the two is better still — so the two regions are carrying partly independent information about the same tumour state. The percentages (28 % vs 41 %) are the **input measurement**; the AUC is the **performance measure**. Judging a biomarker on the input rather than the performance is the error option E is built on.

Option by option — why each one stands or falls.

A.	is right — exon 2 separates better on its own (AUC 0.924 vs 0.847) and the two regions combine to discriminate best of all (AUC 0.938), so the choice is made on performance, not on how much methylation there is. Full reasoning above.

B.	fails because the course says location matters for diagnostic value; it does not rank the two regions by their distance from the transcription start site. And the mechanism of silencing in the course is **promoter CpG hypermethylation**, not exon 2 methylation acting as a silencer.
C.	fails because it is contradicted by the numbers: combining raises AUC from 0.924 to 0.938, so the regions are not redundant.
D.	fails because it over-corrects the adjacent-tissue caveat. The caveat is that adjacent tissue is *not* the same as truly normal tissue, so it cannot be labelled "healthy" — which changes what your **reference group** must be. It does not make the tumour-versus-adjacent contrast unusable; it means you need a real healthy comparator before you can claim population screening.
E.	fails because quoting the mean difference while ignoring spread is the classic biomarker error. With SDs of 132–166 ng/mL against a mean gap of roughly 95 ng/mL, the distributions overlap heavily, so many individuals fall on the wrong side of any cut-off chosen from means alone. A cut-off needs an ROC, and clinical use needs sensitivity and specificity at a chosen threshold.

In systems terms. AUC is the ROC of the classifier: one number for ranking quality across all operating points. The mean difference is comparing two centroids while ignoring the spread — it says the averages differ, not that a *new, unseen* point can be classified. Combining exon 1 and exon 2 is ensembling two features: partial independence, so the ensemble scores better than either feature alone.

Q4 — Answer: **B**

Why B is right. The two designs differ in **how many inputs their gate has**.

- **Design 2 (synNotch)** is a genuine **two-input AND gate**: the CAR gene is expressed only when the tumour antigen *and* a second contextual ligand are both present. Normal tissue that expresses the antigen alone cannot fire it, because the second input never arrives. That is the whole point of the circuit — it changes the logic from OR to AND, and it is why specificity rises sharply.
- **Design 1 (HRE-controlled CAR)** is a **single-input gate**: low oxygen is the only condition it reads. Its selectivity therefore rests entirely on the assumption that the healthy tissue you care about is not hypoxic. The course's own figure puts hypoxic tissue in roughly half of studied solid tumours — so "hypoxic" is a property of the tumour microenvironment, not a tumour identity. Any normal tissue that is hypoxic (ischaemic, inflamed, metabolically stressed) sits in the same trigger zone.

Option by option — why each one stands or falls.

A.	fails because Design 1's gate reads only one variable. The antigen is not part of its condition at all, so it is not an AND gate.
B.	is right — synNotch is a two-input AND gate while the HRE-controlled CAR is a single-input gate whose selectivity rests on the assumption that healthy tissue is not hypoxic, and hypoxia is a property of the microenvironment rather than an identity. Full reasoning above.
C.	fails because hypoxia is not tumour-exclusive, and the course's figure says nothing of the sort — it says half of solid tumours have it, which is a statement about tumours, not a statement that normal tissue never does.
D.	fails because it inverts the circuit. The whole design intent of synNotch is that *either* input alone is **insufficient**.
E.	fails as an over-claim. The course presents both as strategies: Design 1 buys selectivity without needing a second antigen, at the price of depending on the oxygen environment; Design 2 buys context-gating at the price of a second marker and a larger construct. Neither dominates everywhere.

In systems terms. Design 1 is a feature flag driven by one environment variable (the deployment stage); Design 2 is a boolean AND over two flags. Design 1's false positives are wherever that variable is set; Design 2's are wherever *both* are set. Widening the input set is a different operation from tightening a threshold.

Q5 — Answer: **A**

Why A is right. The course's cross-talk chain is **hypomethylation of the HDAC promoter → ↑ HDAC → ↓ histone acetylation → ↓ transcription**. Read it as a circuit: HDAC is a *repressor* of the permissive chromatin state, so more HDAC means less acetylation means less transcription. An HDAC inhibitor acts on the **acetylation node**, which is downstream of the methylation node — so it can restore acetylation and re-express a silenced tumour suppressor. But it does nothing to the methyl groups themselves: removing CpG methylation is the job of the DNMT/TET axis, not of an HDAC inhibitor. And because the active state needs **both** loss of DNA methylation **and** gain of acetylation, restoring one of them while the other is still wrong leaves the locus short of fully open.

Option by option — why each one stands or falls.

A.	is right — the drug acts on the acetylation node downstream of the methylation node, so it can restore acetylation and re-express a silenced tumour suppressor, but it cannot remove the CpG methyl groups, and the active state needs both. Full reasoning above.
B.	fails because the sign is wrong twice over. Following the course's own chain in reverse, hypermethylating the HDAC promoter would *lower* HDAC, which would *raise* acetylation and *raise* transcription. And the general claim — that methylation at a promoter always increases expression — is backwards: promoter hypermethylation suppresses.
C.	fails because this is the over-generalisation the course's chain exists to defeat. The sign of the methylation → expression relation depends on **what the gene does**. Hypomethylating the HDAC promoter lowers a *repressor* and therefore raises transcription of the target; hypomethylating a tumour-suppressor promoter would do the opposite. "Hypomethylation = activation" is not a law.
D.	fails because it is false. HDACs act on histones — that is why HDAC inhibitors work at all and restore silenced tumour suppressors. p53, HSP90, NF-κB and tubulin are *additional* non-histone substrates, not the only ones.
E.	fails because it contains two errors. The chain is **negative** feedback (more HDAC → less acetylation → less transcription of the HDAC gene → less HDAC), so it stabilises rather than runs away; and a positive-feedback loop of this kind is exactly what HDAC inhibitors are designed to break.

In systems terms. This is a thermostat with inverted polarity. The **sign of the gain** decides whether the loop stabilises the system or runs away from it — the classic control-theory distinction. Reading a negative-feedback regulator as a runaway amplifier is the same class of error as treating a thermostat as a heater.

PART B — Pipeline mapping: model answers

What a full-mark answer contains, stage by stage, with the marks each limb carries and the trap each part is built around.

Q6 — Model answer (6 marks)

(a) The full pipeline, stage by stage (2 marks). Name the technique, its input, its output, and one thing the stage cannot establish on its own.

1	**Sample handling / material choice.** Input: tumour tissue and patient serum, with its handling history recorded. Output: a sample whose upstream is known. *Cannot establish:* whether the analyte survived the journey — that only shows up later, and it is why the first decision is a decision about the material.
2	**Discovery proteomics — 2-DE.** Input: lysed tumour tissue. Output: a two- dimensional map of spots — first dimension **isoelectric focusing** on a pH 4.0–9.0 strip (charge / pI), second dimension **SDS-PAGE** from 220 kDa down to 10 kDa (molecular weight). *Cannot establish:* which protein any given spot is. A spot is a coordinate, not a name: separation and identification are two separate components of biomarker discovery.
3	**Protein identification — LC-MS/MS.** Input: the excised spot. Output: the protein identity — the spot is excised, digested (trypsin), the peptides separated by LC, then fragmented in a second MS stage down to **two-amino-acid fragments**, from which the sequence is assembled **de novo**. *Cannot establish:* that the protein is made by the tumour cells rather than by stroma or infiltrating immune cells, or that it is secreted at all. (Plain LC-MS would only give a peptide mass fingerprint that must match a database, so it could not have named this protein at all.)
4	**Validation in tissue.** Input: the candidate plus the same tissue. Output: the candidate confirmed as expressed and attributable to tumour cells — by **qPCR** (transcript), **IHC** (protein and its location), **Western blot** (size and quantity). *Cannot establish:* whether the protein can be measured in blood.

5	**Blood-based assay — ELISA.** Input: serum (or plasma, chosen and stated deliberately). Output: a quantified concentration in ng/mL per patient. *Cannot establish:* how the assay performs in the population it is meant for — a test on already-diagnosed patients is not a screening test until it has been run on the right cohort.
6	**Cohort evaluation and the statistical decision.** Input: the labelled cohort — disease-free controls, early disease, later disease, with the matrix fixed. Output: an ROC curve, an AUC, a chosen threshold with its sensitivity and specificity, and a stated claim about what the test can and cannot detect. *This is the stage that licenses clinical use*, and it is the only stage that can tell you whether the assay can screen.

(b) Where FFPE fails (1 mark). It fails at **stage 1**, before the pipeline starts, and it fails silently. FFPE preserves architecture for histology and IHC but crosslinks and degrades nucleic acid: the RNA profile is a smear with no clean 28S and 18S peaks. Every later stage inherits the damage — fewer and weaker 2-DE spots, unreliable identifications by LC-MS/MS, a qPCR/IHC/Western signal that reflects degradation rather than biology, an ELISA whose concentrations measure what survived the tissue processing, and finally an ROC built on noise. The rule is the course's own: **the same material cannot be right for both jobs** — FFPE for histology, fresh frozen for profiling.

(c) Why ≤ 2 cm decides the screening claim, and what the SDs imply (1 mark). A screening test is judged on **early** disease, because by the time a tumour is > 2 cm it is often already symptomatic — a large signal there buys nothing. Note also that the > 2 cm mean (231.3 ng/mL) is *not* higher than the ≤ 2 cm mean (271.5 ng/mL), so the marker is not simply tracking tumour bulk; it tracks early disease specifically. The standard deviations (61.3 for controls; 165.9 and 132.3 in the HCC groups) are very large relative to the mean gap of roughly 95 ng/mL between controls and ≤ 2 cm disease. The distributions therefore overlap heavily: many individual patients fall on the wrong side of any cut-off, which means **poor specificity** and a real false-positive burden in screening. The assay needs an ROC, a defined threshold and reported sensitivity and specificity before it can be offered to a population.

(d) The comparator group you need (1 mark). You need a **genuinely disease-free group** — healthy volunteers confirmed to have no disease — alongside the tumour cohort, with the same sample matrix and the same assay. Tumour-adjacent tissue is **biochemically different from truly normal tissue**, so it is a valid comparator for showing a difference is *tumour-associated*, but it cannot be labelled "healthy" and it cannot establish that the marker is absent in the general population. The adjacent- tissue comparison alone licenses a claim about the tumour; it does not license a claim about screening the public.

(e) Classification (1 mark). As measured here, Vimentin is a **diagnostic** biomarker: the data are cross-sectional and separate diseased from non-diseased subjects, which is what detection and early detection require. To become **prognostic** it needs follow-up outcome data — survival or recurrence — and a survival analysis (Kaplan–Meier by marker status) to show it predicts outcome regardless of treatment. To become **predictive** it needs a treatment-interaction dataset: patients on a defined therapy, marker status against response, showing the marker predicts differential benefit from that therapy. Neither conversion is possible from the present data, because the data contain no outcome variable and no treatment variable at all.

Q7 — Model answer (6 marks)

(a) The pipeline, stage by stage (2 marks).

1	**Collection.** Input: the tumour, the tumour-infiltrating lymphocytes (TIL) and the patient's blood. Output: fresh tissue and fresh cells.
2	**T-cell isolation.** Input: TIL from the tumour and PBMC from blood. Output: a purified T-cell population free of the other cells in the sample.
3	**Activation.** Input: the isolated T cells. Output: viable, transcriptionally active cells that will expand — see (d) below.
4	**Single-cell sequencing, restricted to the TCR.** Input: the activated cells. Output: one transcriptome per cell, plus the rearranged TCR sequence of each cell. Whole-genome sequencing is unnecessary here — the library can be restricted to the TCR genes.
5	**Clone calling.** Input: the per-cell TCR sequences. Output: the expanded clones — cells sharing a TCR come from one activated cell that proliferated, so the clones that grew disproportionately are the ones with an anti-tumour response.
6	**Antigen inference.** Input: the TCR sequence. Output: a computationally predicted peptide — the neoantigen the receptor should bind.
7	**Antigen capture by the independent route.** Input: tumour cells carrying MHC–peptide complexes. Output: the peptides the tumour actually presents — immunopeptidomics: pull the complexes down, digest, MS/MS, de novo sequence.
8	**Merge and shortlist.** Input: the prediction and the presented peptides. Output: **20–30 candidate peptides** that survive both lines of evidence.

9	**In vitro validation.** Input: candidate peptides plus the patient's PBMC/T cells. Output: **IFN-γ ELISpot** — the number of antigen-specific cells secreting IFN-γ — with **flow cytometry** for the CD8 population and activation markers. See (c).
10	**Scale-up.** Input: validated epitopes. Output: a **personalised** vaccine for one patient, or the identification of **shared** epitopes for an off-the-shelf vaccine. See (e).

(b) The two independent routes, and why two (1 mark).

- **Route 1 — TCR-seq → prediction.** It starts from the receptor: it tells you **which clone expanded**, and therefore which antigen mattered, but the antigen itself is a *computational inference* from sequence similarity and may simply be wrong.
- **Route 2 — immunopeptidomics → MS/MS de novo.** It starts from the tumour: it tells you **what is actually presented** on the tumour's MHC, with no prediction step, but it cannot say which clone recognises it.

They fail in **different ways**, which is exactly why they are worth having. A prediction can be wrong; a presented peptide can be irrelevant (presented but not recognised, or presented by an HLA context the clone does not read). Only when the predicted peptide is *also* found presented — and then shown to activate the clone's own T cells — do you hold the antigen. Two independent measurements converging on the same peptide is a stronger claim than one measurement plus its own echo.

(c) The validation step that cannot be skipped (1 mark). In vitro peptide stimulation with IFN-γ ELISpot, supported by flow cytometry for CD8 and activation markers. ELISpot counts the antigen-specific cells that secrete IFN-γ (ELISA, by contrast, measures the concentration of soluble protein — they answer different questions). Prediction alone cannot license manufacture, because a predicted peptide may not be processed in that patient, may not be presented in that HLA context, or may be presented without being reactive. The assay establishes the functional link **antigen → T-cell response**, and nothing upstream of it can substitute for it.

(d) Why anti-CD3 plus anti-CD28 (1 mark). They deliver the two signals that T-cell activation requires: **anti-CD3 gives signal 1** (engagement of the T-cell receptor), and **anti-CD28 gives signal 2** (co-stimulation). Signal 1 alone tends to produce anergy rather than activation; signal 2 is what maintains survival and permits proliferation. That matters here because the readout of the whole experiment is *which clones expanded* — and clones only expand if the cells were genuinely activated and kept alive. (The same anti-CD3/CD28 beads with IL-2 and IL-15 do this in the CAR-T manufacturing workflow.)

(e) Personalised versus off-the-shelf (1 mark). The deciding question is whether the **epitope is private or public**.

- **Personalised** — the vaccine is built from that patient's own mutations, because the epitope is **patient-specific**: present in that patient's tumour and absent from controls and from a cohort. To justify the label you must show the epitope is private **and** that the patient's own T cells respond to it (validated ELISpot).
- **Off-the-shelf** — the vaccine uses **shared epitopes** found by sequencing a large cohort of the same cancer type, because the epitope recurs across many patients. To justify the label you must show the epitope is shared across a substantial fraction of the cohort **and** that the HLA context is compatible across them, so one product works for many people.

PART C — Exception handling: model answer

Where the system breaks, what that does to every observation in the table, and how you would prove where the break is.

Q8 — Model answer (8 marks)

(a) The pipeline, and exactly where it breaks (3 marks).

The pipeline, in order:

translation of the CAR mRNA → signal-peptide-directed entry into the rough ER → folding by chaperones and N-linked glycosylation → vesicle to the Golgi (cis → cisternae → trans) → secretory vesicle → plasma membrane → CAR displayed (scFv extracellular, transmembrane anchor, CD3ζ intracellular) → HER2 binding → CD3ζ ITAM signalling through ZAP70 → activation, IL-2, proliferation, perforin/granzymes → tumour cell death.

The break is at the first step: ER entry. With the signal peptide deleted, the nascent chain is made on free cytosolic ribosomes and never enters the ER. Now follow the consequences, and match them to the table:

- **No ER entry → no chaperone-assisted folding, no glycosylation.** The cytosolic copy is misfolded and is degraded, so whole-cell lysate shows **less** CAR protein than the parental construct — exactly the table's Western blot.
- **No ER → no Golgi → no secretory vesicle → no surface CAR.** This is the flow cytometry result: CAR not detected on the cell surface.
- **The transmembrane domain cannot rescue this.** It is an *anchor* for a protein that has already been delivered to the membrane; it is not a *transport signal*, and it cannot import a cytosolic protein into the ER. With the anchor but no route, there is nothing to anchor.
- **CAR mRNA is normal**, because the defect is post-transcriptional — the mRNA is made and translated, and the protein simply never becomes a working receptor. This matches the qPCR row.
- **The cells transduce and survive**, because the CAR is not needed for their own survival; the loss only matters when they meet a target.
- **The scFv binds HER2 when produced in E. coli** — bacteria have no ER and need no signal peptide to fold a small domain — so the antigen-recognition function is intact. The failure is in localisation and trafficking, not in recognition.
- **No surface CAR → no antigen engagement → no CD3ζ signalling → no activation, no proliferation, no cytotoxicity → no killing of HER2-positive targets.** That is the functional assay row, and it is a *downstream consequence*, not a second fault.

(b) Why the postdoc is wrong (1 mark). The signal peptide and the transmembrane domain do different jobs. The **transmembrane domain is an anchor** — it holds a protein in the membrane once it is there. The **signal peptide is an address label** that sends the nascent chain into the ER, where folding, glycosylation and quality control happen, and from where the Golgi delivers the protein to the cell surface. Without ER entry the protein never reaches the Golgi and never reaches the membrane, so the receptor is simply absent — the cell is left with an anchor and nothing to anchor.

(c) The two most likely failure points, and the experiment that separates them (2 marks).

1	**ER entry itself failed** — the protein is synthesised in the cytosol, cannot fold, and is degraded; nothing reaches the surface. *Discriminating experiment:* **surface versus intracellular FACS with the same anti-scFv (or anti-Fc) antibody, with and without permeabilisation.** If surface staining is negative but intracellular staining is also weak, while the lysate Western shows only a faint band, the protein is being made and destroyed in the cytosol → failure point 1.
2	**Trafficking or surface delivery failed after ER entry** — the protein reaches the ER and folds, but is retained in the ER or early Golgi and never reaches the plasma membrane (this is the residual possibility if the deletion left a weak or cryptic signal peptide, or if another trafficking step is compromised). *Same experiment:* surface-negative but intracellular-strong staining → the protein is made and folded but not delivered → failure point 2.

A supporting assay for the same split: **confocal microscopy or intracellular staining with an ER marker (calnexin) and a Golgi marker (GM130)** to see where the retained protein sits — cytosolic, ER, or Golgi. An N-glycosidase-sensitive Western blot or a lectin blot answers the same question more bluntly: **if the protein is glycosylated, it entered the ER.**

(d) A class of therapeutic protein the same defect abolishes (1 mark). Any **secreted protein** — a therapeutic cytokine (IL-2), an enzyme replacement (factor VIII), or a monoclonal antibody. Each requires ER entry for folding and glycosylation and exits through the Golgi and secretory vesicles; without the signal peptide they are made in the cytosol, do not fold, and are **never secreted**. The product is not merely mislocalised — it is absent. Antibody–drug conjugates fail the same way: the antibody needs the ER and the Golgi both to reach the cell surface and to be internalised after binding.

(e) The general rule (1 mark). The signal peptide is a routing requirement, not an accessory. Any protein that must enter the secretory pathway cannot be rescued by carrying a membrane anchor of its own: removing the routing signal does not degrade the output, it deletes it. And because localisation sits upstream of everything — folding, glycosylation, trafficking, display, and therefore function — no downstream fix can recover a protein that never reached the ER. In pipeline terms: the routing step feeds every stage after it, so if it fails, every later stage has no input, and the first thing to verify in any expression experiment is *where the protein is*, not how much of it was made.

Self-marking scale

Band	What it looks like
25–30	You are answering from mechanism, not recall. Chains are correct and the limits are named.
20–24	Solid on Parts A and B; the weak spot is usually Part C — you found the fault but stopped before following it downstream.
15–19	You know the content but not the wiring. Re-do the two pipeline questions and draw the chain before writing.
Below 15	Work the Lecture 6 chains (Chains 2, 3, 5, 6, 7, 9) and the case bank, then sit this paper again.

What happens next

Submit your answers and the grading is done properly: a full post-mortem, question by question, with every biological error translated into computational or data-flow terms — what the mistake would look like if the mechanism were code, and which part of the pipeline your reasoning broke at.

Source basis: `GNBF5020_Complete_Revision_All_Lectures.md` *(Lectures 1–5, audio + slides),* `GNBF5020_Lecture_6_Exam_Focus_and_Learning_Guide.md` *(Lecture 6),* `GNBF5020_Case_Studies.md`*,* `GNBF5020_Exam_Focus_Prediction.md`*. The question paper is* `GNBF5020_Practice_Exam_01_Cumulative_L1–L6.pdf`*.*